

HOSPITAL MANAGEMENT SYSTEM

Database Design and SQL Implementation Report

1. Abstract

Hospitals are vital institutions providing medical facilities for diverse health conditions. This system addresses the complexities of manual record-keeping by implementing a structured relational database. By automating day-to-day activities such as doctor scheduling, patient diagnosis, and medical history management, the system significantly reduces manual effort, eliminates operational errors, and ensures a seamless flow of healthcare activities.

2. System Overview

The Hospital Management System is a comprehensive SQL-based solution designed to optimize healthcare operations. The database architecture consists of seven core entities that facilitate efficient data storage, retrieval, and manipulation.

Primary Entities:

- **Physician:** Stores details of medical practitioners and their specializations.
- **Department:** Manages various hospital wings and administrative heads.
- **Nurse:** Tracks nursing staff and their registration status.
- **Patient:** Contains demographic data and primary care assignments.
- **Procedures:** Catalogs medical services, imaging tests, and standardized costs.

3. Database Schema Definitions

3.1 Table: Physician

Field	Type	Null	Key	Description
EmployeeID	Int	NO	PRI	Unique ID for the doctor
Name	Varchar(150)	NO		Full name

Field	Type	Null	Key	Description
Position	Varchar(150)	NO		Job Title/Specialty

3.2 Table: Patient

Field	Type	Null	Key	Extra
Patient_ID	Int	NO	PRI	Auto Increment
Name	Varchar(100)	NO		First Name
Surname	Varchar(100)	NO		Last Name
Gender	Varchar(150)	NO		Demographic data
Primary_Check	Int	NO	MUL	FK to Physician

4. SQL Implementation Samples

```
-- Database Initialization
CREATE DATABASE HOSPITAL_MANAGEMENT_SYSTEM;
USE HOSPITAL_MANAGEMENT_SYSTEM;

-- Table Creation Example (Physician)
CREATE TABLE Physician (
    employeeid INT PRIMARY KEY,
    name VARCHAR(150) NOT NULL,
    position VARCHAR(150) NOT NULL
);

-- Data Insertion Example
INSERT INTO Physician (employeeid, name, position) VALUES
(1, 'Dr. John Dorian', 'Staff Internist'),
(2, 'Dr. Elliot Reid', 'Attending Physician'),
(3, 'Dr. Christopher Turk', 'Surgical Attending Physician');
```

5. Advanced Data Retrieval

Patient & Physician Relationship

Retrieving patients along with the name of the physician who performed their primary check-up:

```
SELECT
    CONCAT(p.name, ' ', p.surname) AS Patient_Name,
    ph.name AS Treating_Physician
FROM Patient p
LEFT JOIN Physician ph ON p.primary_check = ph.employeeid;
```

Medical Cost Analysis

Query to identify procedures that cost more than the hospital average:

```
SELECT name, cost
FROM procedures
WHERE cost > (SELECT AVG(cost) FROM procedures);
```

6. Conclusion

The Hospital Management System digitizes critical medical workflows, providing a robust framework for managing patient care and administrative tasks. The relational schema ensures data integrity while SQL queries provide actionable insights into hospital operations. This implementation serves as a foundational step toward a fully integrated digital healthcare environment.